

giving

$$\mathcal{E}_o(\mathbf{r}) = -iAz_o \left[\frac{e^{ikz}}{z} e^{ik\frac{x^2+y^2}{2z}} \right] e^{-\frac{x^2+y^2}{w^2(z)}} \quad (6.59)$$

where we recognize the bracketed term as a spherical wave centered at $z = 0$. Thus far from the beam waist we have a Gaussian spherical wave (but remember this is true not too far from the z -axis).

Beam quality M^2

The propagation of a Gaussian beam is completely described by its waist and its divergence. For an ideal TEM₀₀ beam, the product of the beam waist, ω_o , and the divergence angle, θ_o , is given by the wavelength divided by π .

$$\omega_o \theta_o = \frac{\lambda}{\pi} \quad (6.60)$$

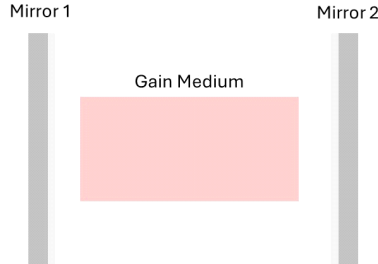
Achievable beams do not follow the ideal diffraction limited performance. When a beam deviates from the ideal Gaussian, the product of the beam waist and the divergence angle increases by the *beam quality factor*, M^2 .

$$\omega \theta = M^2 \frac{\lambda}{\pi} \quad (6.61)$$

The factor M^2 represents how many times wider the focused spot is that the theoretical minimum. An M^2 of 2, for example, means that the focused beam will be twice the ideal minimum spot size. High quality beams may achieve $M^2 < 1.1$. Multimode lasers produce poorer beam qualities. A typical value may be $M^2 = 4$.

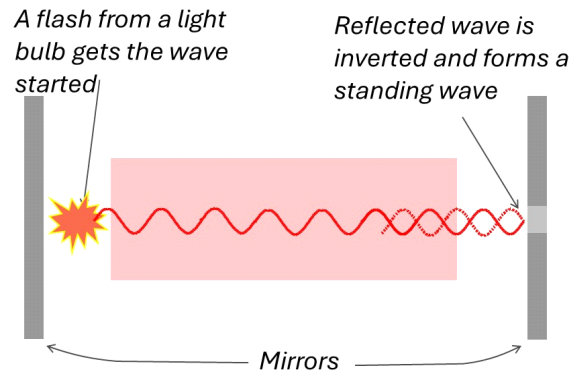
6.2 Laser Resonators

In designing a laser we need the light to travel through the gain medium multiple times. A simple method of achieving this is to put our gain medium in between two parallel mirrors.

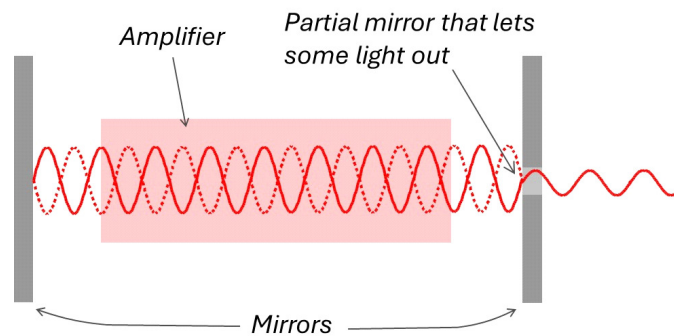


This effectively creates what we will call a *cavity* in which the light will bounce back and forth in one direction.

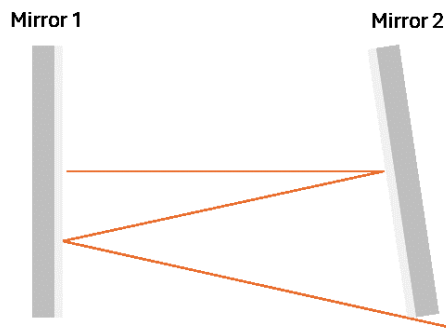
The idea is that we will form a standing wave in our cavity. A flash of light or a diode beam will start the wave



Because we have a cavity, the standing wave will form from reflections

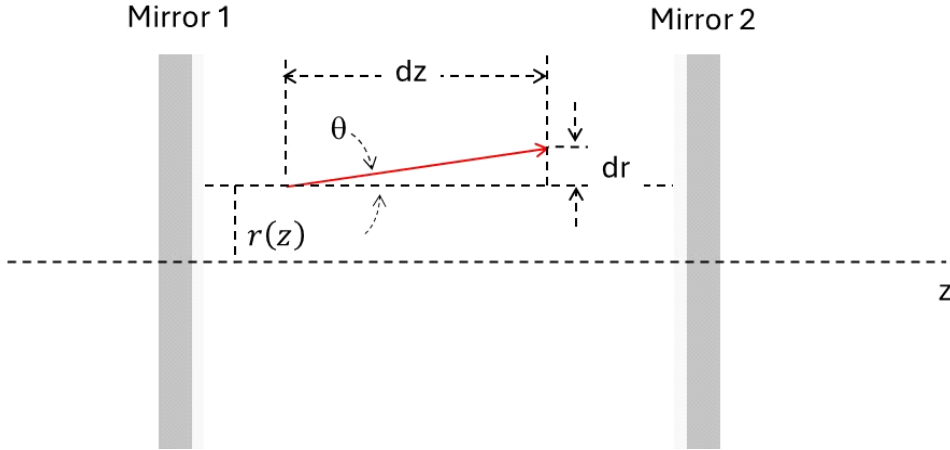


The standing wave is important because we want the light to pass through the gain medium many times and be amplified each time. But this simple design has its problems. If the mirrors are misaligned, the standing wave may not form because the light will leave the cavity.



It will take some work to understand the waves in a cavity in detail, but we can

look at the stability of a cavity to see if the light will leave due to misalignment using geometric optics. In the geometric optics approximation light travels as rays drawn like lines with arrow heads to show which way the light is going. We will find that we want our rays to go mostly in the z -direction. We could think of a z -axis going down the middle of our laser cavity and then consider rays that might not be perfectly aligned with that axis.



6.2.1 ABCD matrices and ray tracing

Let's think of such a ray that is a distance $r(z)$ from the z -axis and has a slope

$$r'(z) = \frac{dr}{dz}$$

(I have exaggerated the size of dr and dz in the figure). Looking at the figure, we can see that

$$\tan \theta = \frac{dr}{dz}$$

and because we are only considering rays that are nearly parallel to the axis (something we will have to justify later), we can say that θ is small.

$$\tan \theta \approx \theta \approx r'(z)$$

Let's for now assume cylindrical symmetry about the z -axis. And let's take the positive direction to be in the direction the beam is propagating.

To find where the ray will go we need a way to calculate the displacement and slope of a ray at a point z_2 knowing the displacement and slope of the ray at point z_1 . If our cavity is empty (vacuum) then there is no (geometrical) reason for the ray to change directions between our two points and we can say that

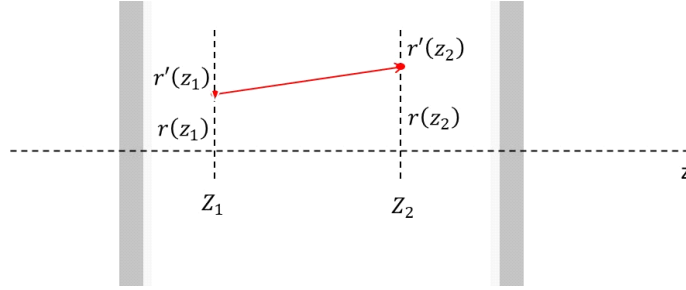
$$r(z_2) = r(z_1) + r'(z_1)(z_2 - z_1)$$

and

$$r'(z_2) = r'(z_1)$$

It is customary to write these two equations as a single matrix equation

$$\begin{bmatrix} r(z_2) \\ r'(z_2) \end{bmatrix} = \begin{bmatrix} 1 & z_2 - z_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r(z_1) \\ r'(z_1) \end{bmatrix}$$



This notation is bulky, we could make it more compact if we called

$$\begin{bmatrix} r(z_1) \\ r'(z_1) \end{bmatrix} = \begin{bmatrix} r_i \\ r'_i \end{bmatrix}$$

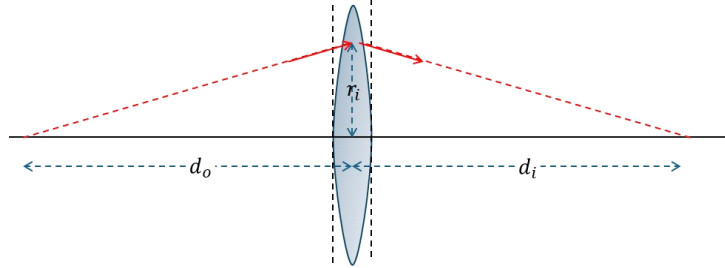
and

$$\begin{bmatrix} r(z_2) \\ r'(z_2) \end{bmatrix} = \begin{bmatrix} r_f \\ r'_f \end{bmatrix}$$

and note that $z_2 - z_1$ is just the distance d between the two points. Then

$$\begin{bmatrix} r_f \\ r'_f \end{bmatrix} = \begin{bmatrix} 1 & d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r_i \\ r'_i \end{bmatrix}$$

This notation can be extended to many different optical elements within the geometric limit. For a thin lens,



if we look just before the lens we have r_i and just after the lens we have r_f and it is clear that the displacement from the axis is just

$$r_i = r_f$$

But the slopes change due to refraction. Suppose we have a point object (source of light) a distance d_o from the lens, and we form an image at a distance d_i from the the lens. Then

$$r'_i = \frac{r_i}{d_o}$$

$$r'_f = -\frac{r_i}{d_i}$$

and we know that for a thin lens

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$$

so if

$$d_o = \frac{r_i}{r'_i}$$

$$d_i = -\frac{r_i}{r'_f}$$

$$\frac{1}{\frac{r_i}{r'_i}} + \frac{1}{-\frac{r_i}{r'_f}} = \frac{1}{f}$$

$$\frac{r'_i}{r_i} - \frac{r'_f}{r_i} = \frac{1}{f}$$

$$r'_i - r'_f = r_i \frac{1}{f}$$

and finally

$$r'_f = r'_i - r_i \frac{1}{f}$$

Using this last equation and the fact that $r_i = r_f$ we can make our matrix equation for the lens

$$\begin{bmatrix} r_f \\ r'_f \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \begin{bmatrix} r_i \\ r'_i \end{bmatrix}$$

But notice that this is for just before the lens to just after the lens. No distance traveled in this equation.

Lenses are useful and so this last equation is important, but we could have done the same thing for a curved spherical mirror. Only for mirrors we know that $f = \frac{R}{2}$

$$\begin{bmatrix} r_f \\ r'_f \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{2}{R} & 1 \end{bmatrix} \begin{bmatrix} r_i \\ r'_i \end{bmatrix}$$

Like with all of geometric optics, there is a sign convention. Let's take

$$\begin{aligned} r' > 0 & \quad r \text{ is increasing with propagation} \\ r' < 0 & \quad r \text{ is decreasing with propagation} \end{aligned}$$

We can generalize this approach. We can take

$$\begin{bmatrix} r_f \\ r'_f \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} r_i \\ r'_i \end{bmatrix}$$

where the matrix

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

is called the ray matrix or more commonly the *ABCD* matrix. There is a different *ABCD* matrix for each type of optical element including (as we have seen) propagation in vacuum.

We can go back to our lens example, but this time let's account for the distance to the lens. That would be

$$\begin{bmatrix} r_{f1} \\ r'_{f1} \end{bmatrix} = \begin{bmatrix} 1 & d_o \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r_i \\ r'_i \end{bmatrix}$$

before the lens. Then just after the lens we would take

$$\begin{bmatrix} r_{f2} \\ r'_{f2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \begin{bmatrix} r_{f1} \\ r'_{f1} \end{bmatrix}$$

We can combine these two equations into one

$$\begin{aligned} \begin{bmatrix} r_{f2} \\ r'_{f2} \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix} \begin{bmatrix} 1 & d_o \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r_i \\ r'_i \end{bmatrix} \\ &= \begin{bmatrix} 1 & d_o \\ -\frac{1}{f} & 1 - \frac{1}{f}d_o \end{bmatrix} \begin{bmatrix} r_i \\ r'_i \end{bmatrix} \end{aligned}$$

Then to go to the image location we need another vacuum propagation.

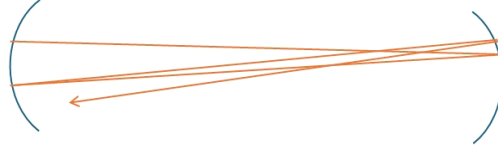
$$\begin{aligned} \begin{bmatrix} r_{f3} \\ r'_{f3} \end{bmatrix} &= \begin{bmatrix} 1 & d_i \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r_{f2} \\ r'_{f2} \end{bmatrix} \\ \begin{bmatrix} r_{f3} \\ r'_{f3} \end{bmatrix} &= \begin{bmatrix} 1 & d_i \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & d_o \\ -\frac{1}{f} & 1 - \frac{1}{f}d_o \end{bmatrix} \begin{bmatrix} r_i \\ r'_i \end{bmatrix} \\ \begin{bmatrix} r_{f3} \\ r'_{f3} \end{bmatrix} &= \begin{bmatrix} 1 - \frac{1}{f}d_i & d_o - d_i\left(\frac{1}{f}d_o - 1\right) \\ -\frac{1}{f} & 1 - \frac{1}{f}d_o \end{bmatrix} \begin{bmatrix} r_i \\ r'_i \end{bmatrix} \end{aligned}$$

All of this is a great way to describe geometrically how light goes through an optical system. You might be able to guess how we could write optical ray tracing software like OSLO or CodeV after thinking about *ABCD* matrices. But for laser design there is another important reason for *ABCD* matrices. We can determine if a laser cavity is stable from the cavity *ABCD* matrix.

6.2.2 Cavity Stability

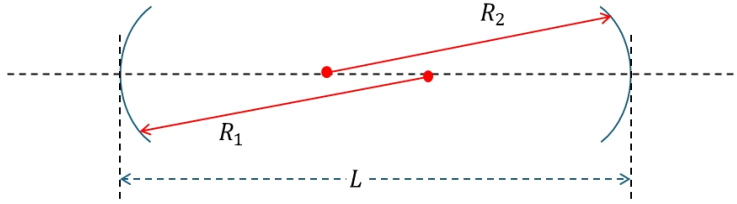
Ray matrices can be used to determine the stability of a laser cavity. Our misaligned mirror example is an example of an unstable laser cavity. The rays eventually leave the cavity. In a stable cavity the rays do not leave.

We found that a cavity made from flat mirrors that were misaligned was unstable. And perfect alignment is hard to do. Here is another design



These curved mirrors are also a little misaligned, but they seem to be doing better in keeping the light in the cavity. We can use our $ABCD$ matrices to find a condition for when a cavity is stable. Let's look at a cavity with curved mirrors in more detail.

The mirrors are separated by a distance L and they have radii of curvature R_1 and R_2 .



Let's start a ray at the left hand mirror and find an $ABCD$ matrix for a round trip back to the mirrors. The ray will travel a distance L , turn at M_2 , and travel a distance L back to M_1 and finally let's turn our ray around again at M_1 . Our $ABCD$ matrix will consist of a combination of each of these items

$$\begin{aligned}
 \begin{bmatrix} A & B \\ C & D \end{bmatrix} &= \begin{bmatrix} 1 & 0 \\ -\frac{2}{R_1} & 1 \end{bmatrix} \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\frac{2}{R_2} & 1 \end{bmatrix} \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 - 2\frac{L}{R_2} & L - L\left(2\frac{L}{R_2} - 1\right) \\ \frac{2}{R_2}\left(2\frac{L}{R_1} - 1\right) - \frac{2}{R_1} & 1 - 2\frac{L}{R_1} - L\left(\frac{2}{R_1} - \frac{2}{R_2}\left(2\frac{L}{R_1} - 1\right)\right) \end{bmatrix} \\
 &= \begin{bmatrix} 1 - 2\frac{L}{R_2} & 2L - 2\frac{L^2}{R_2} \\ 4\frac{L}{R_1 R_2} - \frac{2}{R_2} - \frac{2}{R_1} & 1 - \frac{2L}{R_2} - \frac{4L}{R_1} + \frac{4L^2}{R_1 R_2} \end{bmatrix}
 \end{aligned}$$

Then after one round trip we have

$$\begin{bmatrix} r_1 \\ r'_1 \end{bmatrix} = \begin{bmatrix} 1 - 2\frac{L}{R_2} & 2L - 2\frac{L^2}{R_2} \\ 4\frac{L}{R_1 R_2} - \frac{2}{R_2} - \frac{2}{R_1} & 1 - \frac{2L}{R_2} - \frac{4L}{R_1} + \frac{4L^2}{R_1 R_2} \end{bmatrix} \begin{bmatrix} r_i \\ r'_i \end{bmatrix}$$

We could let our ray go through N round trips. Then we would have

$$\begin{bmatrix} r_N \\ r'_N \end{bmatrix} = \begin{bmatrix} 1 - 2\frac{L}{R_2} & 2L - 2\frac{L^2}{R_2} \\ 4\frac{L}{R_1 R_2} - \frac{2}{R_2} - \frac{2}{R_1} & 1 - \frac{2L}{R_2} - \frac{4L}{R_1} + \frac{4L^2}{R_1 R_2} \end{bmatrix}^N \begin{bmatrix} r_i \\ r'_i \end{bmatrix}$$

Let's write this again as just another ABCD matrix

$$\begin{bmatrix} r_N \\ r'_N \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}^N \begin{bmatrix} r_i \\ r'_i \end{bmatrix}$$

and note that the ABCD matrix has a determinant of

$$AD - CB = 1$$

This isn't obvious. But let's go back to the parts of our ABCD matrix.

$$\begin{vmatrix} 1 & L \\ 0 & 1 \end{vmatrix} = 1$$

$$\begin{vmatrix} 1 & 0 \\ -\frac{2}{R_2} & 1 \end{vmatrix} = 1$$

we can see that the determinant of each part of our ABCD matrix is 1 and from linear algebra we know that the determinant of the product of matrices is equal to the product of the determinants. Then

$$\begin{vmatrix} A & B \\ C & D \end{vmatrix} = 1 \times 1 \times 1 \times 1 = 1$$

We want to use this $ABCD$ matrix to determine how stable the cavity is. To do that let's use an approach developed by a researcher named Sylvester. It's not obvious. So let's take it a step at a time.

Let's define an angle θ using

$$\cos(\theta) = \frac{1}{2}(A + D)$$

It's not yet clear what this angle means. But let's do it just the same. Then using this odd angle definition we can use Sylvester's theorem. We take

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{N+1} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix}^N$$

And we can guess a form for $\begin{bmatrix} A & B \\ C & D \end{bmatrix}^N$ using our odd definition of θ

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^N = \frac{1}{\sin \theta} \begin{bmatrix} A \sin(N\theta) - \sin((N-1)\theta) & B \sin(N\theta) \\ C \sin(N\theta) & D \sin(N\theta) - \sin((N-1)\theta) \end{bmatrix}$$

This is certainly not obvious. But if we assume it is true then

$$\begin{aligned}
& \begin{bmatrix} A & B \\ C & D \end{bmatrix}^{N+1} \\
&= \frac{1}{\sin \theta} \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} A \sin(N\theta) - \sin((N-1)\theta) & B \sin(N\theta) \\ C \sin(N\theta) & D \sin(N\theta) - \sin((N-1)\theta) \end{bmatrix} \\
&= \frac{1}{\sin \theta} \begin{bmatrix} BC \sin N\theta - A(\sin \theta(N-1) - A \sin N\theta) & AB \sin N\theta - B(\sin \theta(N-1) - D \sin N\theta) \\ CD \sin N\theta - C(\sin \theta(N-1) - A \sin N\theta) & BC \sin N\theta - D(\sin \theta(N-1) - D \sin N\theta) \end{bmatrix} \\
&= \frac{1}{\sin \theta} \begin{bmatrix} BC \sin N\theta - A \sin \theta(N-1) + A^2 \sin N\theta & AB \sin N\theta - B \sin \theta(N-1) + BD \sin N\theta \\ CD \sin N\theta - C \sin \theta(N-1) + CA \sin N\theta & BC \sin N\theta - D \sin \theta(N-1) + D^2 \sin N\theta \end{bmatrix} \\
&= \frac{1}{\sin \theta} \begin{bmatrix} BC \sin N\theta + A^2 \sin N\theta - A \sin \theta(N-1) & AB \sin N\theta + BD \sin N\theta - B \sin \theta(N-1) \\ CD \sin N\theta + CA \sin N\theta - C \sin \theta(N-1) & BC \sin N\theta + D^2 \sin N\theta - D \sin \theta(N-1) \end{bmatrix} \\
&= \frac{1}{\sin \theta} \begin{bmatrix} (BC + A^2) \sin N\theta - A \sin \theta(N-1) & (AB + BD) \sin N\theta - B \sin \theta(N-1) \\ C(D + A) \sin N\theta - C \sin \theta(N-1) & (BC + D^2) \sin N\theta - D \sin \theta(N-1) \end{bmatrix}
\end{aligned}$$

This is quite a mess. Let's look at just one element, the 1, 1 element

$$(BC + A^2) \sin N\theta - A \sin \theta(N-1)$$

since

$$AD - CB = 1$$

$$CB = AD - 1$$

and then

$$(BC + A^2) \sin N\theta - A \sin \theta(N-1)$$

becomes

$$\begin{aligned}
& (AD - 1 + A^2) \sin N\theta - A \sin \theta(N-1) \\
& (AD + A^2 - 1) \sin N\theta - A \sin \theta(N-1) \\
& (AD + A^2) \sin N\theta - \sin N\theta - A \sin \theta(N-1) \\
& A(D + A) \sin N\theta - \sin N\theta - A \sin \theta(N-1)
\end{aligned}$$

And since

$$\cos(\theta) = \frac{1}{2}(A + D)$$

then

$$(A + D) = 2 \cos \theta$$

so

$$\begin{aligned}
& A(2 \cos \theta) \sin N\theta - \sin N\theta - A \sin \theta(N-1) \\
& 2A \sin N\theta \cos \theta - \sin N\theta - A \sin \theta(N-1)
\end{aligned}$$

Now we need a trig identity (though not one I remembered!)

$$\sin N\theta \cos \theta = \frac{1}{2} \sin(N+1)\theta + \frac{1}{2} \sin(N-1)\theta$$

Using this we have

$$\begin{aligned}
 & 2A \left(\frac{1}{2} \sin(N+1)\theta + \frac{1}{2} \sin(N-1)\theta \right) - \sin N\theta - A \sin(N-1)\theta \\
 & A \sin(N+1)\theta + A \sin(N-1)\theta - \sin N\theta - A \sin(N-1)\theta \\
 & A \sin(N+1)\theta - \sin N\theta
 \end{aligned}$$

Which was quite a trip. So far we have

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{N+1} = \frac{1}{\sin \theta} \begin{bmatrix} A \sin(N+1)\theta - \sin N\theta & (AB + BD) \sin N\theta - B \sin \theta (N-1) \\ C(D + A) \sin N\theta - C \sin \theta (N-1) & (BC + D^2) \sin N\theta - D \sin \theta (N-1) \end{bmatrix}$$

and not much looks any clearer. But let's try the 1,2 element

$$\begin{aligned}
 & (AB + BD) \sin N\theta - B \sin \theta (N-1) \\
 & B(A + D) \sin N\theta - B \sin \theta (N-1) \\
 & B2 \cos \theta \sin N\theta - B \sin \theta (N-1) \\
 & 2B \left(\frac{1}{2} \sin(N+1)\theta + \frac{1}{2} \sin(N-1)\theta \right) - B \sin \theta (N-1) \\
 & B \sin(N+1)\theta + B \sin(N-1)\theta - B \sin \theta (N-1) \\
 & B \sin(N+1)\theta
 \end{aligned}$$

and

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{N+1} = \frac{1}{\sin \theta} \begin{bmatrix} A \sin(N+1)\theta - \sin N\theta & B \sin(N+1)\theta \\ C(D + A) \sin N\theta - C \sin \theta (N-1) & (BC + D^2) \sin N\theta - D \sin \theta (N-1) \end{bmatrix}$$

We have two more to go. The 2,1 element is

$$\begin{aligned}
 & C(D + A) \sin N\theta - C \sin \theta (N-1) \\
 & C2 \cos \theta \sin N\theta - C \sin \theta (N-1) \\
 & 2C \left(\frac{1}{2} \sin(N+1)\theta + \frac{1}{2} \sin(N-1)\theta \right) - C \sin \theta (N-1) \\
 & C \sin(N+1)\theta + C \sin(N-1)\theta - C \sin \theta (N-1) \\
 & C \sin(N+1)\theta
 \end{aligned}$$

so that

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{N+1} = \frac{1}{\sin \theta} \begin{bmatrix} A \sin(N+1)\theta - \sin N\theta & B \sin(N+1)\theta \\ C \sin(N+1)\theta & (BC + D^2) \sin N\theta - D \sin \theta (N-1) \end{bmatrix}$$

And finally, let's do the 2,2 term

$$(BC + D^2) \sin N\theta - D \sin \theta (N-1)$$

since

$$AD - CB = 1$$

then

$$CB = AD - 1$$

and our 2, 2 term is

$$(AD - 1 + D^2) \sin N\theta - D \sin \theta (N - 1)$$

or

$$\begin{aligned} & (AD + D^2 - 1) \sin N\theta - D \sin \theta (N - 1) \\ & (AD + D^2) \sin N\theta - \sin N\theta - D \sin \theta (N - 1) \\ & D(A + D) \sin N\theta - \sin N\theta - D \sin \theta (N - 1) \\ & D(2 \cos \theta) \sin N\theta - \sin N\theta - D \sin \theta (N - 1) \\ & 2D \left(\frac{1}{2} \sin(N + 1)\theta + \frac{1}{2} \sin(N - 1)\theta \right) - \sin N\theta - D \sin \theta (N - 1) \\ & D \sin(N + 1)\theta + D \sin(N - 1)\theta - \sin N\theta - D \sin \theta (N - 1) \\ & D \sin(N + 1)\theta - \sin N\theta \end{aligned}$$

which yields

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{N+1} = \frac{1}{\sin \theta} \begin{bmatrix} A \sin(N + 1)\theta - \sin N\theta & B \sin(N + 1)\theta \\ C \sin(N + 1)\theta & D \sin(N + 1)\theta - \sin N\theta \end{bmatrix}$$

But this is just the equation we guessed at the start with N replaced by $N + 1$.

So our guess for $\begin{bmatrix} A & B \\ C & D \end{bmatrix}^N$ is true for $N + 1$. And if it is true for any $N + 1$ then it must be true for any N and we have proved Sylvester's theorem. Since it works, we can use our odd guess.

$$\begin{bmatrix} r_N \\ r'_N \end{bmatrix} = \frac{1}{\sin \theta} \begin{bmatrix} A \sin(N\theta) - \sin((N - 1)\theta) & B \sin(N\theta) \\ C \sin(N\theta) & D \sin(N\theta) - \sin((N - 1)\theta) \end{bmatrix} \begin{bmatrix} r_i \\ r'_i \end{bmatrix}$$

Some thought might show that for $\begin{bmatrix} r_N \\ r'_N \end{bmatrix}$ to stay finite this angle θ must be real.

and we remember that

$$\cos(\theta) = \frac{1}{2}(A + D)$$

and for one trip across the cavity and back we remember that

$$\begin{bmatrix} r_N \\ r'_N \end{bmatrix} = \begin{bmatrix} 1 - 2\frac{L}{R_2} & 2L - 2\frac{L^2}{R_2} \\ 4\frac{L}{R_1 R_2} - \frac{2}{R_2} - \frac{2}{R_1} & 1 - \frac{2L}{R_2} - \frac{4L}{R_1} + \frac{4L^2}{R_1 R_2} \end{bmatrix}^N \begin{bmatrix} r_i \\ r'_i \end{bmatrix}$$

well now our angle must be

$$\begin{aligned}\cos(\theta) &= \frac{1}{2} \left(\left(1 - 2\frac{L}{R_2} \right) + \left(1 - \frac{2L}{R_2} - \frac{4L}{R_1} + \frac{4L^2}{R_1 R_2} \right) \right) \\ &= \frac{1}{2} \left(2 - 2\frac{L}{R_2} - \frac{2L}{R_2} - \frac{4L}{R_1} + \frac{4L^2}{R_1 R_2} \right) \\ &= 1 - \frac{2L}{R_2} - \frac{2L}{R_1} + \frac{2L^2}{R_1 R_2}\end{aligned}$$

so we can test to see if this makes θ real.

It might be easier (since we have $\cos \theta$) to make sure

$$|\cos \theta| \leq 1$$

$$-1 \leq 1 - \frac{2L}{R_2} - \frac{2L}{R_1} + \frac{2L^2}{R_1 R_2} \leq 1$$

We could move the lone 1 to make this

$$-2 \leq -\frac{2L}{R_2} - \frac{2L}{R_1} + \frac{2L^2}{R_1 R_2} \leq 0$$

and we can divide by 2

$$-1 \leq -\frac{L}{R_2} - \frac{L}{R_1} + \frac{L^2}{R_1 R_2} \leq 0$$

and bring the 1 back in

$$0 \leq 1 - \frac{L}{R_2} - \frac{L}{R_1} + \frac{L^2}{R_1 R_2} \leq 1$$

For historical reasons we usually define

$$g_1 = 1 - \frac{L}{R_1}$$

and

$$g_2 = 1 - \frac{L}{R_2}$$

then

$$\begin{aligned}g_1 g_2 &= \left(1 - \frac{L}{R_1} \right) \left(1 - \frac{L}{R_2} \right) \\ &= \frac{L^2}{R_1 R_2} - \frac{L}{R_2} - \frac{L}{R_1} + 1\end{aligned}$$

so our condition for stability is

$$0 \leq g_1 g_2 \leq 1$$

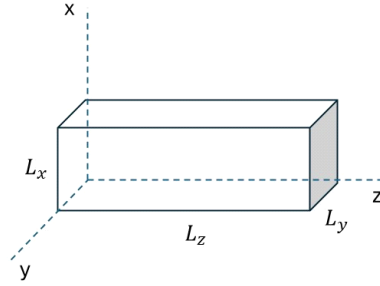
The g_1 and g_2 are called the g parameters of the resonator. This allows us to check for stability without resorting to ray tracing software and trial and error. We can see that whether or not a cavity is stable depends on the radii of curvature of the mirrors and on the length of the cavity.

This was a good geometric view of laser cavities. But we know that our light is really an electromagnetic wave and that it will experience wave phenomena like diffraction. Let's look into the idea of a cavity electromagnetically and see if we can improve on this design.

6.3 Light in a cavity

The important thing in making the standing wave is the size and shape of the cavity. We would like just one standing wave to form, so our laser produces just one wavelength of light. Let's look at an empty cavity and see how many standing wave modes we will get. Remember, each mode may give a different frequency, and we would prefer just one frequency.

Starting with a simple design let's consider an empty cavity that has no gain medium but does have sides to it.



Since the cavity is empty, we can use the no-source maxwell equations to find the fields inside.

$$\nabla \cdot \vec{\mathbf{D}} = 0 \quad (6.62)$$

$$\nabla \cdot \vec{\mathbf{B}} = 0 \quad (6.63)$$

$$\nabla \times \vec{\mathbf{E}} = -\frac{d\vec{\mathbf{B}}}{dt} \quad (6.64)$$

$$\nabla \times \vec{\mathbf{H}} = \frac{\partial \vec{\mathbf{D}}}{\partial t} \quad (6.65)$$

where

$$\vec{\mathbf{B}} = \mu_o \vec{\mathbf{H}}$$

because we have only non-magnetic media in the cavity (really we have only open space with no atoms at all!) and the electric displacement field is given by

$$\vec{\mathbf{D}} = \epsilon_o \vec{\mathbf{E}} + \vec{\mathbf{P}}$$